Ping-Ju Lin

Research Area: Deep Learning (AI), Neuroscience, Biomedical signal processing

pjlin14@gmail.com | homepage | +1 (213) 728-0744

Education

Tsinghua University, Beijing 08/2020-06/2024

Doctor of Philosophy, Mechanical Engineering, Department of Mechanical Engineering

- Advisor: Prof. Linhong Ji; Research Area: Rehabilitation engineering, Intelligent and Biomimetic Machinery

Harvard Medical School and Mass General Hospital, MA 09/2022-03/2023

Visiting Scholar, Department of Neurology, laboratory for Translational Neurorecovery

- Advisor: Prof. David Lin; Research Area: Neurorecovery, Neurology, Neuroscience

George Washington University, DC 09/2018-06/2020

Master of Science, Electrical Engineering, School of Engineering & Applied Science

- Advisor: Prof. Murray H. Loew; Research Area: Medical Image & Image Analysis

Feng Chia University, Taiwan 09/2014-06/2018

Bachelor of Engineering, Electrical Engineering, Department of Electrical Engineering

Publication (Selected)

1 **P.-J. Lin**., Z. Jiang., Y. Liu., W. Li., … Arthur W. Toga., " A Vision-Language Foundation Model with Multi-Teacher Knowledge Distillation for Alzheimer's Disease Diagnosis" in Alzheimer's & Dementia，21, 12, e71029

2 **P. -J. Lin** et al., "Explainable Deep-Learning Prediction for Brain–Computer Interfaces Supported Lower Extremity Motor Gains Based on Multistate Fusion," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 1546-1555, 2024

3 **P. -J. Lin** et al., "AM-EEGNet An Advanced Multi-input deep learning framework for EEG-based classification tasks," *Neurocomputing*, vol. 585, 2024

4 **P. -J. Lin** et al., "A transferable deep learning prognosis model for predicting stroke patients' recovery in different rehabilitation trainings," in *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 12, pp. 6003-6011, 2022

5 **P. -J. Lin** et al., "CNN-Based Prognosis of BCI Rehabilitation Using EEG From First Session BCI Training," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 1936-1943, 2021

6 J. Sun, **P.-J. Lin** et al., "Multimodal behavioral data predict stroke patient's response to BCI treatment through explainable AI," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2025

7 Li, W., Luo F., Liu Y., Zou Y., Mo L., He Q., **Lin, P. J.,** Xu Q., Liu A., Zhang C., Cheng J., Cheng L and Ji L. "Bioinspired Smart Triboelectric Soft Pneumatic Actuator-Enabled Hand Rehabilitation Robot." *Advanced Materials,* 2419059, 2025.

8 J. Sun, T. Jia, **P.-J. Lin**, Z. Li, L. Ji and C. Li, "Multiscale Canonical Coherence for Functional Corticomuscular Coupling Analysis," in *IEEE Journal of Biomedical and Health Informatics,* vol. 28, pp. 812-822, 2023

9 Li, W., Li, C., Liu, A., **Lin, P. J.**, Mo, L., Zhao, H., Xu., Q., Meng., X., Ji, L. "Lesion-specific cortical activation following sensory stimulation in patients with subacute stroke. " *Journal of NeuroEngineering and Rehabilitation*. vol. 20, no.155, 2023

10 Li, Zhibin., Li, Wei., **Lin, Ping-Ju**., Jia, Tianyu., Ji, Linhong., Li, Chong. "Motor-Respiratory Coupling Improves Endurance Performance during Rhythmic Isometric Handgrip Exercise. *Medicine & Science in Sports & Exercise,* 2023.

11 Qian C, Li W, Jia T, Li C, **Lin PJ**, Yang Y, Ji L. Quantitative Assessment of Motor Function by an End-Effector Upper Limb Rehabilitation Robot Based on Admittance Control. *Applied Sciences*. vol 11, pp. 6854, 2021

Patent

1 **P.-J. Lin**. 2022. A method or computer equipment for patients to prognosticate rehabilitation outcome from CN Patent. ZL 2021 1 0593146.1, filed May 28, 2021, and issued August 26, 2022

Research Experiment

Peking University 05/2026-Present

- School of Psychology and Cognitive Science
  1. Interpretable Multimodal Fatigue Biomarker Discovery for Driving Fatigue Assessment
     • Investigated the physiological validity of commonly used driving-fatigue “gold-standard” labels (e.g., PERCLOS, reaction time, and physician EEG annotations) using multimodal neurophysiological signals and self-supervised learning frameworks.
     • Developed an interpretable cross-dataset fatigue representation framework integrating EEG, EOG, ECG, and respiratory signals to discover latent fatigue states, construct continuous fatigue indices, and evaluate cross-dataset generalization across heterogeneous driving-fatigue datasets.

University of Southern California 08/2024-11/2025

- Laboratory of Neuro Imaging
  1. ADLIP: Connecting Medical reports and Neuroimage for Alzheimer’s Disease Patient
     • Developed a contrastive learning model to analyze and predict Alzheimer’s disease progression by integrating medical reports and neuroimaging data.
     • Investigated the relationship between clinical indicators from medical reports and neuroimaging biomarkers (e.g., MRI, PET scans) to identify early signs of Alzheimer’s disease.
     • Evaluated the model’s performance in connecting multimodal data (textual medical reports and neuroimages) to improve the accuracy of Alzheimer’s disease prognosis and patient stratification.
  2. Large Multi-model explanation
     • Utilized deep learning explanation methods (e.g., Saliency Map, Grad-CAM) to visualize critical brain regions and features contributing to Alzheimer’s disease diagnosis and progression.

Tsinghua University 08/2020-06/2024

- Lab of Intelligent and Bio-mimetic Machinery
  1. Brain-Computer Interface upper-limb rehabilitation outcome prediction
     • Developed a convolutional-based model namely the Two-Way CNN model to prognosis upper limb disability patient motor ability gain from the brain-computer interface rehabilitation method.
     • Compared the relationship between the motor recovery and different state signals including the motor state (Event related discrimination) and resting state.
     • Analyzed the model by using a deep learning explanation method (Saliency Map) to visualize the weighing region contributes to a good recovery.
  2. Understanding the feature relate with the recovery of Robotic treatment
     • Developed a quantitative assessment method to better understand the recovery of motor function with the use of robotic rehabilitation systems.
     • Found that movement trajectory was closely related to the patient's motor recovery rate.
  3. Advanced Multi-inputs Deep learning architecture
     • Developed a multi-input deep learning model architecture based on the EEG feature to improve the prediction accuracy in all kinds of the EEG-based prediction problems.
  4. Brain EEG feature explanation with motor recovery
     • Analyze the EEG feature from the deep learning model explanation method to visualize the region and frequency related to stroke patients’ motor recovery.

Tsinghua Chang Geng Hospital 09/2021-06/2024

- Medical Research Center
  1. Transfer Learning in different rehabilitation treatment
     • Developed a deep learning model for stroke cause lower extremity disability patients in predicting two different rehabilitation motor ability gain.
     • Validated the potential of transferring a pre-train model in different rehabilitation methods i.e. one robotic and one traditional method.
     • Found the same recovery pattern for post-stroke patients from the power spectrum density and functional connectivity in two different rehabilitation treatments.
  2. Explainable deep-learning prediction for BCI-supported lower extremity motor gains
     • Collaborate with medical doctors to undergo post-stroke patients with lower extremity disability to treatment with a specific experimental design.
     • Found the critical factor of brain electrical activity related to the motor recovery of lower limbs stroke causes disability patients.

Research Experiment

Massachusetts General Hospital 09/2022-06/2023

Laboratory for Translational Neurorecovery

1. Post-Stroke motor recovery prediction based on Magnetic Resonance Imaging (MRI)

    • Developed three different deep learning architectures (CNN, MLP, transformer) on the MRI dataset to predict the recovery gain after three-month rehabilitation training.
    • Analyzed the model using a deep learning explanation method (SHAP value) to visualize how the weighing region relates to motor recovery.

Industry Experiment

National Information Security Standardization Technical Committee 06/2021-12/2021

- Project Consultant
  1. Advised the standard in EEG pre-processing procedure. (Brain-Computer Interface)
  2. Completed a chapter in the “White paper on the application of brain-computer interface technology in the medical and health field”

Foxconn (Hon Hai Precision Industry) 06/2019-08/2019

- AI/ML Engineer Intern
  1. Developed an NLP (Nature language Processing) model to train the dialect language (Min-Nan) model with the data from the Pi-li company. The goal is to alternate the character's voices.
  2. Programmed a Web Crawler to grip data from social media such as PTT, Dcard, and Facebook and visualize the public opinion poll.

Teaching Experiment

Tsinghua University - RWTH Aachen University Joint Program 2022-2023

- Master Thesis Supervisor
  1. Supervised RWTH Aachen University master’s students on research projects involving experimental design, signal processing, and machine learning methodologies.
  2. Guided students through thesis development, including literature review, data analysis, model development, and scientific writing.

The George Washington University Spring 2020

- Teaching Assistant (Electrical Energy Conversion)
  1. Prepared weekly quizzes and graded weekly quizzes.
  2. Solved individual student problems in-class projects.

Award

- Ministry of Education Natioanal Scholarship for PhD (Best honor) 2022
- Ministry of Education Natioanal Scholarship for PhD (2nd) 2021
- Ministry of Education Natioanal Scholarship for PhD (3nd) 2020

Skills

- Profession knowledge in Deep Learning, Brain-computer interface, and Neuroscience; Skilled in Brain signal processing (EEG, MRI, fNIRS), Deep Learning model architecture in stroke patient recovery, and Stroke patient motor recovery prediction.
- Software: Python, Tensorflow, Keras, Pytorch, MATLAB, EEGLab, C++, ZEMAX, Layout, Verilog, Altera, Quartus
- Languages: Chinese (Native), English (Fluent)